

DATA 308: Lab 7b

Mutiple linear regression

Suppose we are fitting the following regression model:

$$\text{Midterm Grade}_i = \beta_0 + \beta_1 \text{Time Spent Studying}_i + \beta_2 \text{Homework Grade}_i + \beta_3 I(\text{First Time Learning Regression}_i = \text{Yes}) + \epsilon_i$$

For each task (numbered 1 - 5) specify the letter corresponding to the appropriate method:

- a. Partial F Hypothesis Test: $H_0 : \beta_2 = \beta_3 = 0$ vs $H_A : \text{at least one is nonzero}$
 - b. Confidence interval for β_1
 - c. Confidence interval for average outcome
 - d. Prediction interval for a single outcome
 - e. Individual Hypothesis Test: $H_0 : \beta_2 = 0$ vs $H_A : \beta_2 \neq 0$
 - f. F Hypothesis Test: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ vs $H_A : \text{at least one is nonzero}$
 - g. Residual Standard Error (RSE)
 - h. Individual Hypothesis Tests: $H_0 : \beta_1 = 0$ vs $H_A : \beta_1 \neq 0$, $H_0 : \beta_2 = 0$ vs $H_A : \beta_2 \neq 0$, and $H_0 : \beta_3 = 0$ vs $H_A : \beta_3 \neq 0$
-
1. _____ I want to assess how closely my model's predicted Midterm Grades match the observed Midterm Grades in my sample.
 2. _____ I want to formally decide if Homework Grade is linearly associated with Midterm Grade.
 3. _____ I want to formally decide if the Homework Grade and First Time Learning Regression variables are linearly associated with Midterm Grade given Time Spent Studying is in the model.
 4. _____ I want to formally decide if at least one of Homework Grade, First Time Learning Regression, and Time Spent Studying are linearly associated to Midterm Grade.
 5. _____ I want to estimate the linear association between Midterm Grade and Time Spent Studying.

Identify when we may want to use each of the remaining methods: